DIGITAL ASSET INVESTMENT RISK, RETURN, AND DIVERSIFICATION

Learning Outcome

- analyze sources of risk, return, and diversification among digital asset investments

Value appreciation of digital assets, such as Bitcoin and Ethereum, has been rapid since inception, most recently fueled by the introduction of more traditional indirect forms of investment. Because cryptocurrencies are a relatively new innovation, their market is subject to rapid price swings, changes, and uncertainty, with most investors considering them alternative investments.

Digital Asset Investment Risks and Returns

Bitcoin and other cryptocurrency values are based solely on asset appreciation, with no underlying cash flows. The market demand for the limited supply of cryptocurrencies is a significant driver of prices. For instance, the supply of Bitcoin is limited to 21 million Bitcoins, by design. For this reason, Bitcoins are sometimes viewed by some investors as the digital version of gold.

Since its launch in 2009, Bitcoin's performance has been characterized by high return, high volatility, and low correlations with traditional asset classes, as seen in Exhibit 9 and Exhibit 10.

Exhibit 9:

Monthly Log-Returns for Bitcoin and Asset Class Benchmarks between January 2011 and January 2022

	Bitcoin in USD	S&P 500 Index Total Return	MSCI World Index Return	Bloomberg Global Aggregate Index
Average	8.84%	1.13%	0.66%	0.16%
Standard deviation	0.32	0.04	0.04	0.01
Coefficient of variation	3.66	3.43	6.09	8.16

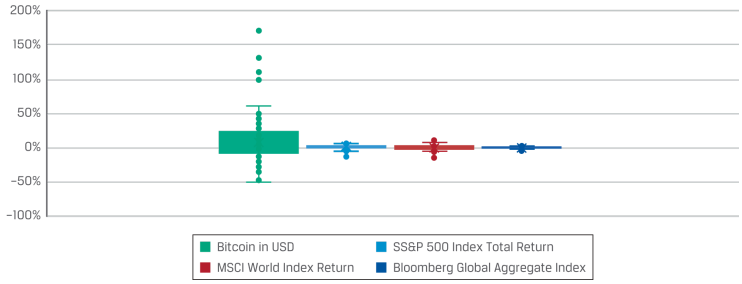
Notes: Bitcoin price in US dollars, S&P 500 Total Return, MSCI World Index, and Bloomberg Global Aggregate Bond Index are log-returns for January 2011 to December 2022. All series are in US dollars.

From mid-2010, when publicly available market prices were available, Bitcoin price has risen from USD0.05 at inception to the historical peak of USD68,789 on 10 November 2021, before a subsequent crash took it down to around USD17,709 on 18 June 2022. Early investors in Bitcoins were hugely rewarded with phenomenal price appreciation.

Investing in cryptocurrencies therefore comes with material risks unique to digital assets. Cryptocurrencies have a relatively brief track record, yet their volatility has remained high, as Exhibit 10 shows. Although the volatility of Bitcoin has declined, it is still much higher than the volatility of traditional financial assets, such as the S&P 500 Index.

Exhibit 10:

January 2022



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Notes: Bitcoin price in US dollars, S&P 500 Total Return, MSCI World Index, and Bloomberg Global Aggregate Bond Index are log-returns for January 2011 to January 2022. All series are in US dollars.

Bitcoin-like price and return patterns are often observed in other cryptocurrencies and may reflect a great deal of uncertainty about cryptocurrencies as an investable asset class. Given that regulation of cryptocurrencies is evolving, there still is no clear legal protection for using them as a medium of exchange. While they are regulated as a digital commodity in the United States and comprehensive rules in the EU remain pending, there is a great deal of uncertainty from an investor perspective about the quality of legal protection an investor can enjoy when investing in these assets. Considering that fraud and criminal activities are rampant among buyers, sellers, issuers, and marketers of digital assets, the continued legal and regulatory uncertainty is non-trivial. Multiple countries have placed extensive restrictions on trading and owning cryptocurrencies, such as China, which banned the asset in 2021.

Diversification Benefits of Digital Asset Investments

Although purely speculative in nature, considered by some to have value drivers distinct from common equity and debt markets, cryptocurrencies have exhibited low correlations with traditional asset class returns. This may provide a hint that the long-term determinants of cryptocurrency prices differ from those of traditional investment assets. In practice, prices (or returns) of cryptocurrencies are driven more by market adoption, network effects, technological advancement, regulatory development, speculation, and general market risk appetite. Some of these factors are unique to this asset class. Exhibit 11 shows the monthly correlation of Bitcoin with some benchmarks.

Exhibit 11:

Correlations between Monthly Log-Returns for Cryptocurrencies and Selected Asset Classes (January 2011 to January 2022)

	Bitcoin in US Dollars	S&P 500 Index Total Return	MSCI World Index Return	Bloomberg Global Aggregate Index
Bitcoin in US dollars	1			
S&P 500 Index Total Return	0.21	1		
MSCI World Index Return	0.22	0.97	1	
Bloomberg Global Aggregate Index	0.14	0.25	0.33	1

Notes: Bitcoin price in US dollars, S&P 500 Total Return, MSCI World Index, and Bloomberg Global Aggregate Bond Index are log-returns for January 2011 to January 2022. All series are in US dollars.

diversification benefits of cryptocurrencies remain to be seen.

Question Set

1. More institutional investors are allocating capital to cryptocurrencies because of their:

   A. low price volatility.
   B. high expected cash flow.
   C. low correlation with other traditional asset classes.

   Hide Solution

   Solution:

   The correct answer is C. The value of Bitcoin and other cryptocurrencies is based solely on asset appreciation, with no underlying cash flows. Historically, their price volatility has been extremely high. As such, cryptocurrencies are often considered a high-risk investment. Nevertheless, more institutional investors allocate their investment into cryptocurrencies because of the low historical correlation with other traditional asset classes, providing potential diversification benefits to a portfolio.

2. Explain the historical characteristics of cryptocurrencies' correlation with other traditional investments.

   Hide Solution

   Solution:

   Cryptocurrencies have historically exhibited low long-term correlations with traditional asset class returns. Prices of cryptocurrencies are driven more by market adoption, network effects, technological advancement, regulatory development, and general market risk appetite. Some of these factors are unique to this asset class. But as institutional participation increases in digital assets, the correlations with traditional assets have risen.

3. Identify the following statement as true or false: Regulatory uncertainty is a non-trivial risk for investors in cryptocurrencies.

   Hide Solution

   Solution:

   True. Regulatory uncertainty is a non-trivial risk for investors in cryptocurrencies because there is considerable risk of fraud and criminal activity among participants, and legal protection over these assets remains uncertain. Regulation of cryptocurrencies is evolving; some countries have adopted them, while some have banned them altogether.

OPTIONAL UNTESTED SEGMENT

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